Trion® T20 BGA256 Development Kit User Guide

T20F256-DK-UG-v1.9
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www.efinixinc.com

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Introduction

Thank you for choosing the Trion® T20 BGA256 Development Kit (part number: T20F256C-DK), which allows you to explore the features of the T20 FPGA.

Warning: The board can be damaged without proper anti-static handling.

What's in the Box?

- Trion® T20 BGA256 Development Board preloaded with a demonstration design
- 4 standoffs
- 4 screws
- 3 foot micro-USB cable (type B)

Note: Shipout device may vary between C4 and I4.

Download the Efinity® Software

To develop your own designs for the T20 device on the board, you must install the Efinity® software. You can obtain the software from the Support Center.

The Efinity® software includes tools to program the device on the board. Refer to the Efinity® Software User Guide for information about how to program the device.

Learn more: Efinity® documentation is installed with the software (see **Help** > **Documentation**) and is also available in the Support Center under Documentation.

Installing the Linux USB Driver

The following instructions explain how to install a USB driver for Linux operating systems.

1. Disconnect your board from your computer.
2. In a terminal, use these commands:

```
> sudo <installation directory>/bin/install_usb_driver.sh
> sudo udevadm control --reload-rules
> sudo udevadm trigger
```

Note: If your board was connected to your computer before you executed these commands, you need to disconnect it, then re-connect it.

Installing the Windows USB Drivers

Note: If you have another Efinix board and are using the Trion® T20 BGA256 Development Board, you must manage drivers accordingly. Refer to **AN 050: Managing Windows Drivers** for more information.

On Windows, you use software from Zadig to install drivers. Download the Zadig software (version 2.7 or later) from zadig.akeo.ie. (You do not need to install it; simply run the downloaded executable.)

To install the driver:

1. Connect the board to your computer with the appropriate cable and power it up.
2. Run the Zadig software.

   Note: To ensure that the USB driver is persistent across user sessions, run the Zadig software as administrator.

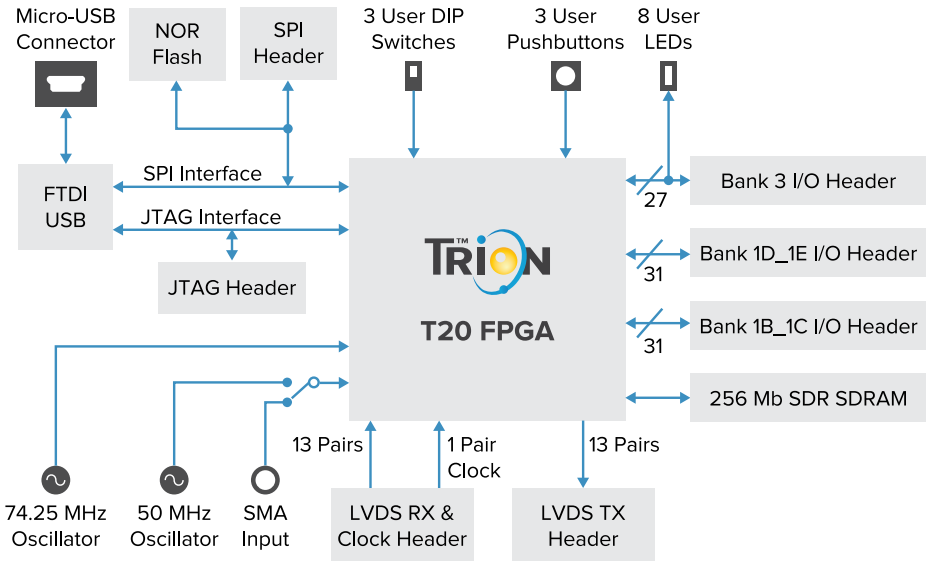
3. Choose **Options > List All Devices**.
4. Repeat the following steps for each interface. The interface names end with *(Interface N)*, where *N* is the channel number.
   - Select **libusb-win32** in the **Driver** drop-down list.
   - Click **Replace Driver**.
5. Close the Zadig software.

Note: This section describes how to install the libusb-win32 driver for each interface separately. If you have previously installed a composite driver or installed using libusbK drivers, you do not need to update or reinstall the driver. They should continue to work correctly.

Board Functional Description

The Trion® T20 BGA256 Development Board contains a variety of components to help you build designs for the Trion® T20 device.

Figure 1: Trion® T20 BGA256 Development Board Block Diagram



Features

- Compact design (106.7 mm x 76.2 mm) (4.2" x 3")
- Efinix® T20F256C device in an 256-ball FineLine BGA package
- FTDI FT2232H dual-channel chipset with USB controller
- Winbond 32 Mbit SPI NOR flash memory
- Micro-USB type AB receptacle
- Power:
  - Power source: 5 V 4 A power supply or USB 5 V, 500 mA USB (for low-power consumption designs)
  - On-board switching regulators (maximum at 2.0 A) source 3.3 V, 2.5 V, and 1.2 V components; one on-board dropout regulator (maximum at 500 mA) sources 1.8 V components
  - Selectable 3.3 V, 2.5 V, and 1.8 V VDDIO for T20F256C I/O banks 1D_1E and 1B_1C
  - Fixed 3.3 V VDDIO for T20F256C I/O banks 1A, 3, and 4
  - 5 V output header to provide power for external devices
- 50 MHz and 74.25 MHz oscillators for T20F256C PLL input
- Optional external clock source available through SMA input to drive the T20F256C PLL input or clock input pin
- User inputs:
  - 8 LEDs on T20F256C bank 3 for user outputs
  - 3 pushbutton switches (connected to bank 1A I/O pins)

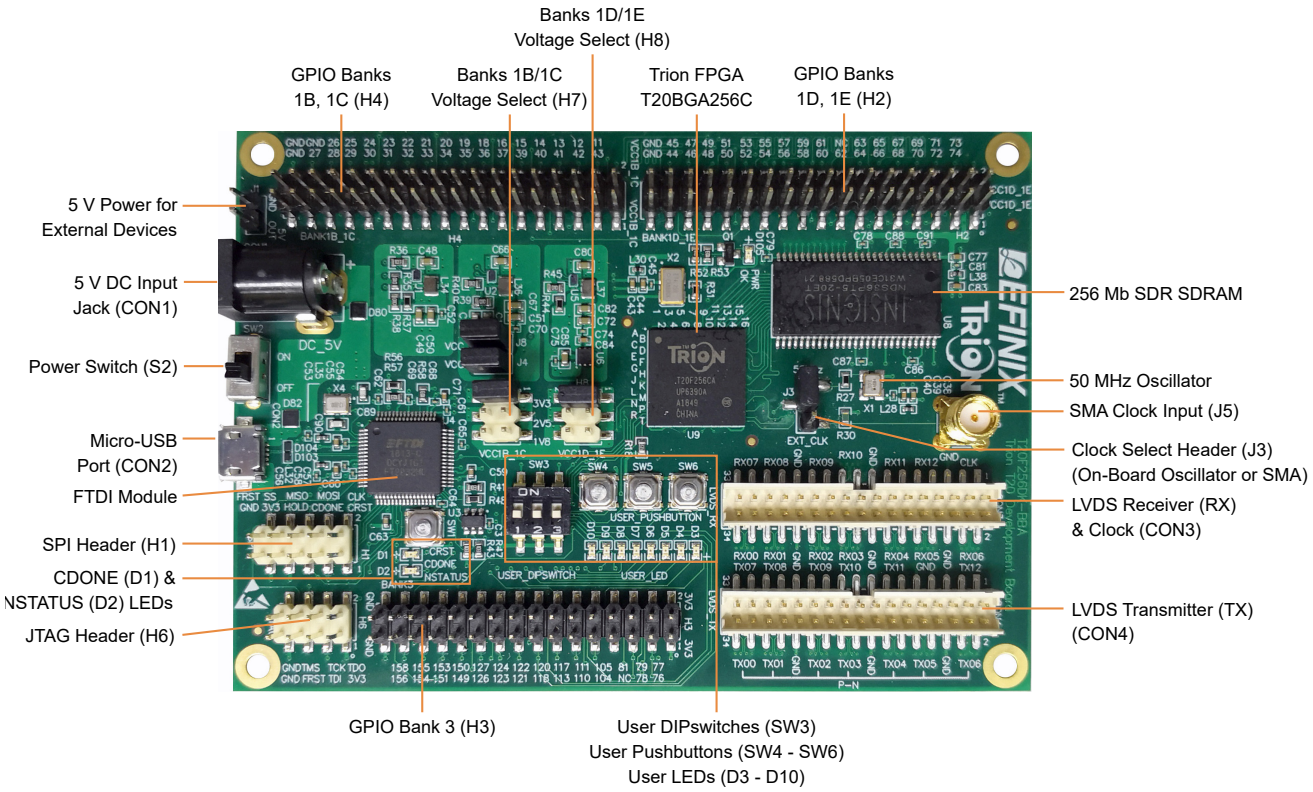
  - 3 DIPswitches (connected to bank 3 I/O pins) for user inputs
- 3 GPIO headers and 2 LVDS headers to connect to external devices
- Power good and T20F256C configuration done LEDs

Overview

The board features the Efinix® T20 programmable device in a 256-ball FBGA package, which is fabricated using Efinix® Quantum® technology. The Quantum®-accelerated programmable logic and routing fabric is wrapped with an I/O interface in a small footprint package. T20 devices also include embedded memory blocks and multiplier blocks (or DSP blocks). You create designs for the T20 device in the Efinity® software, and then download the resulting configuration bitstream to the board using the USB connection.

Learn more: For more information on T20 FPGAs, refer to the T20 Data Sheet.

Figure 2: Trion® T20 BGA256 Development Board Components



The FTDI FT2232H module has two channels to support SPI (FTDI interface 0) and JTAG (FTDI interface 1) configuration. It receives the T20 configuration bitstream from a USB host and writes to the on-board SPI NOR flash memory. After a reset in SPI passive mode, the FTDI controller can also write the configuration bitstream directly to the FPGA. Additionally, it supports direct JTAG programming mode in which it writes the configuration bitstream directly to the FPGA through the JTAG interface.

Learn more: Refer to AN 006 Configuring Trion FPGAs for more information.

The SPI NOR flash memory stores the configuration bitstream it receives from the FTDI FT2232H module. The T20 device accesses this configuration bitstream when it is in active configuration mode (default).

The SDRAM device provides 256 Mb of memory and has a 16-bit data bus with 4 banks.

Note: Although the Trion® T20 BGA256 Development Board has a different power-up sequence, you should follow the power-up sequence in the T20 Data Sheet when designing your own board. For improved reliability, Efinix® recommends that you use supervisor IC at CRESET_N explained in AN 006 Configuring Trion FPGAs.

The board's main power supply is the 5 V DC input. You must use your own DC power supply to provide the board with power through the 5 V input jack. The recommended power input is a 5 V (4 A maximum) DC power source. You can also power the board through the micro USB port for designs with low power consumption (< 500 mA).

The board regulates down the 5 V DC input using on-board switching regulators to provide the necessary voltages for the T20 device, SPI flash memory, SDRAM and on-board oscillator.

Learn more: Depending on your board revision, refer to either the Trion T20 BGA256 (Rev. BBA) Development Board Schematics and BOM or Trion T20 BGA256 (Rev. 3) Development Board Schematics and BOM for more information about the components used in the Trion® T20 BGA256 Development Board.

Power On

To turn on the development board, turn on switch SW2. Upon power-up, the 5 V DC power supply or micro-USB power is input to the on-board regulators through 5 V input jack (CON1) to generate the required 3.3 V, 2.5 V, 1.8 V, and 1.2 V for components on the board. When these voltages are up and stable, the board asserts a **PWR OK** signal (pulled high) from the components' respective regulators. When the board asserts the **PWR OK** signal, a green LED (D105) turns on, giving you a visual confirmation that the power supplies on the board are up and stable.

Note: The micro-USB power supply powers up the board with limited current. Efinix recommends that you use an external DC 5 V DC supply if your user design requires high power.

Reset

The T20F256C device is typically brought out of reset with the CRESET signal. Upon power up, the T20F256C device is held in reset until CRESET toggles high-low-high.

Note: You can manually assert the high-low-high transition with pushbutton switch SW1.

CRESET has a pull-up resistor. When you press SW1, the board drives CRESET low; when you release SW1, the board drives CRESET high. Thus, a single press of SW1 provides the required high-low-high transition.

After toggling CRESET, the T20F256C device goes into configuration mode and reads the device configuration bitstream from the flash memory. When configuration completes successfully, the device drives the CDONE signal high. CDONE is connected to a green LED (D1), which turns on when the T20F256C device enters user mode.

Clock Sources

Two on-board oscillators, 50 MHz and 74.25 MHz, are available to drive the T20F256C PLL input pin and clock input. Alternatively, you can disable the 50 MHz oscillator and use an external clock source through the SMA input (J5). Set jumper J3 to use the 50 MHz or SMA input as the clock source.

Clock Source	PLL Input Pin	Clock Input Pin
50 MHz oscillator or SMA input	GPIOR_157_PLLIN	GPIOR_125_CLK10
74.25 MHz oscillator	GPIOL_75_PLLIN1	–

You can supply a clock to the PLL or clock network in the FPGA through a board header. Refer to H2, H3, and H4 under Headers on page 8 for the dedicated clock pins.

Headers

The board contains a variety of headers to provide power, inputs, and outputs, and to communicate with external devices or boards.

Table 1: Trion® T20 BGA256 Development Board Headers

Reference Designator	Description
CON1	5 V DC power supply input jack
CON2	Micro-USB Type-AB receptacle
CON3	34-pin header for LVDS receiver (RX) and LVDS receiver clock (CLK)
CON4	34-pin header for LVDS transmitter (TX)
H1	SPI header
H2	36-pin header for bank 1D and 1E I/O
H3	36-pin header for bank 3 I/O
H4	36-pin header for bank 1B and 1C I/O
H6	JTAG header
H7	Selects 3.3 V, 2.5 V, or 1.8 V power for banks 1B and 1C
H8	Selects 3.3 V, 2.5 V, or 1.8 V power for banks 1D and 1E
J1	2-pin header for 5 V output
J3	3-pin header to select whether to use the on-board 50 MHz oscillator or SMA input from external clock source
J5	SMA connector for external 3.3 V clock source input

Header CON1 (5 V Power)

CON1 is a 5 V DC power supply input jack. CON1 supplies power to regulators on the board that power the T20F256C FPGA. The maximum current supply to this input jack is 4 A.

Header CON2 (USB Power)

CON2, a micro-USB type B socket, is the interface between the board and your computer for power and communication. Connect the micro-USB cable for configuring T20F256C FPGA and NOR flash. The board supports three different configuration modes: SPI passive mode, SPI active mode, and JTAG mode. The USB cable provides a maximum of 500 mA.

Headers CON3 and CON4 (LVDS)

CON3 and CON4 contain the LVDS signals. CON3 has 13 dedicated LVDS RX channels, and one dedicated LVDS RX clock. CON4 has 13 dedicated LVDS TX channels. You can use LVDS pins for use as GPIO.

Learn more: Refer to the Trion Interfaces User Guide for instructions on use the LVDS pins as GPIO.

Figure 3: LVDS Headers CON3 and CON4

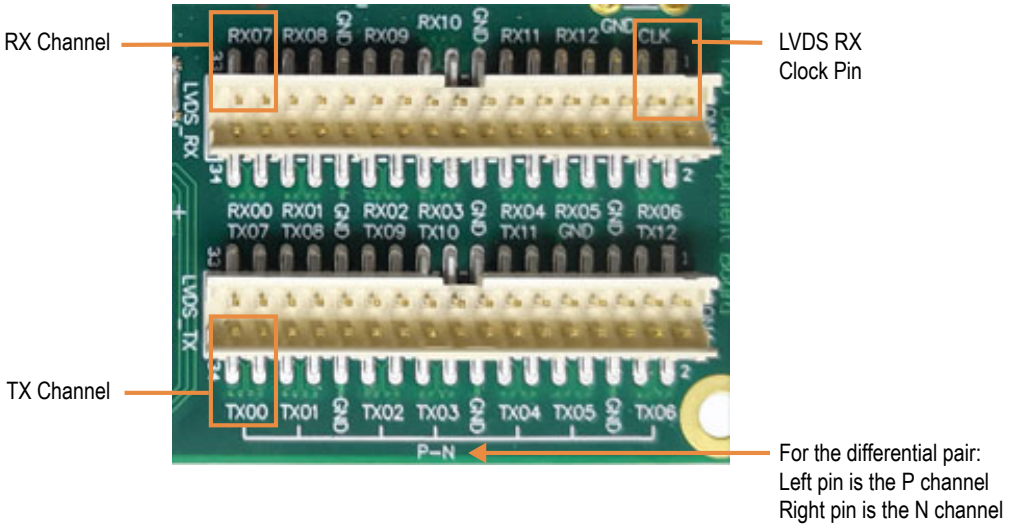


Table 2: CON3 Pin Assignments

Pin Number	T20F256C Pin Name	Description	Pin Number	T20F256C Pin Name	Description
1	GPIOB_CLKN0	Dedicated LVDS RX clock	2	GPIOB_RXN06	Dedicated LVDS RX Channel 6
3	GPIOB_CLKP0		4	GPIOB_RXP06	
5	GND	Ground	6	GND	Ground
7	GPIOB_RXN12	Dedicated LVDS RX Channel 12	8	GPIOB_RXN05	Dedicated LVDS RX Channel 5
9	GPIOB_RXP12		10	GPIOB_RXP05	
11	GPIOB_RXN11	Dedicated LVDS RX Channel 11	12	GPIOB_RXN04	Dedicated LVDS RX Channel 4
13	GPIOB_RXP11		14	GPIOB_RXP04	
15	GND	Ground	16	GND	Ground
17	GPIOB_RXN10	Dedicated LVDS RX Channel 10	18	GPIOB_RXN03	Dedicated LVDS RX Channel 3
19	GPIOB_RXP10		20	GPIOB_RXP03	
21	GPIOB_RXN09	Dedicated LVDS RX Channel 9	22	GPIOB_RXN02	Dedicated LVDS RX Channel 2
23	GPIOB_RXP09		24	GPIOB_RXP02	
25	GND	Ground	26	GND	Ground
27	GPIOB_RXN08	Dedicated LVDS RX Channel 8	28	GPIOB_RXN01	Dedicated LVDS RX Channel 1
29	GPIOB_RXP08		30	GPIOB_RXP01	
31	GPIOB_RXN07	Dedicated LVDS RX Channel 7	32	GPIOB_RXN00	Dedicated LVDS RX Channel 0
33	GPIOB_RXP07		34	GPIOB_RXP00	

Table 3: CON4 Pin Assignments

Pin Number	T20F256C Pin Name	Description	Pin Number	T20F256C Pin Name	Description
1	GPIOB_TXN12	Dedicated LVDS TX Channel 12	2	GPIOB_TXN06	Dedicated LVDS TX Channel 6
3	GPIOB_TXP12		4	GPIOB_TXP06	
5	GND	Ground	6	GND	Ground
7	GND		8	GPIOB_TXN05	Dedicated LVDS TX Channel 5
9	GND		10	GPIOB_TXP05	
11	GPIOB_TXN11	Dedicated LVDS TX Channel 10	12	GPIOB_TXN04	Dedicated LVDS TX Channel 4
13	GPIOB_TXP11		14	GPIOB_TXP04	
15	GND	Ground	16	GND	Ground
17	GPIOB_TXN10	Dedicated LVDS TX Channel 9	18	GPIOB_TXN03	Dedicated LVDS TX Channel 3
19	GPIOB_TXP10		20	GPIOB_TXP03	
21	GPIOB_TXN09	Dedicated LVDS TX Channel 8	22	GPIOB_TXN02	Dedicated LVDS TX Channel 2
23	GPIOB_TXP09		24	GPIOB_TXP02	
25	GND	Ground	26	GND	Ground
27	GPIOB_TXN08	Dedicated LVDS TX Channel 7	28	GPIOB_TXN01	Dedicated LVDS TX Channel 1
29	GPIOB_TXP08		30	GPIOB_TXP01	
31	GPIOB_TXN07	Dedicated LVDS TX Channel 6	32	GPIOB_TXN00	Dedicated LVDS TX Channel 0
33	GPIOB_TXP07		34	GPIOB_TXP00	

Header H1 (SPI)

H1 is a SPI interface that you can use to configure the on-board NOR flash or T20F256C FPGA.

Table 4: H1 Pin Assignment

Pin Number	Signal Name	Description	T20F256C Pin Name
1	CRESET_B	Configuration reset pin (active low)	CRESET_N
2	SPI_CLK	SPI configuration clock	GPIOL_01_CCK
3	CDONE	Configuration done status pin	CONDONE
4	SPI_MOSI	SPI serial data output	GPIOL_08_CDI0
5	HOLD	SPI hold pin (active low)	-
6	SPI_MISO	SPI serial data input	GPIOL_09_CDI1
7	3V3	3.3 V power supply	-
8	SPI_SS	SPI slave select pin (active low)	GPIOL_00_SS
9	GND	Ground	-
10	FT_RST	Reset pin for on board FTDI FT2232 chipset (active low)	-

Headers H2, H3, and H4 (GPIO)

The board headers H2, H3, and H4 contain the Trion T20 Development Board GPIO pins.

- H2 (36 pins) links to bank 1D and 1E GPIO pins. You select the VCCIO with the pins on header H8. Refer to **Headers H7 and H8** on page 15 for details.
- H3 (32 pins) links to bank 3 GPIO pins. VCCIO is fixed at 3.3 V.
- H4 (36 pins) links to bank 1B and 1C GPIO pins. You select the VCCIO with the pins on header H7. Refer to **Headers H7 and H8** on page 15 for details.

Table 5: H2 Pin Assignments

Pin Number	T20F256C Pin Name	DDIO Mode Supported	Pin Number	T20F256C Pin Name	DDIO Mode Supported
1	VCCIO1D_1E	-	2	VCCIO1D_1E	-
3	GPIOL_74(1)	No	4	GPIOL_73	No
5	GPIOL_72	No	6	GPIOL_71	No
7	GPIOL_70(2)	No	8	GPIOL_69(2)	No
9	GPIOL_68	No	10	GPIOL_67	No
11	GPIOL_66	No	12	GPIOL_65	No
13	GPIOL_64	No	14	GPIOL_63	No
15	GPIOL_62	No	16	-	-
17	GPIOL_60	Yes	18	GPIOL_61	Yes
19	GPIOL_58	Yes	20	GPIOL_59	Yes
21	GPIOL_56	Yes	22	GPIOL_57	Yes
23	GPIOL_54	Yes	24	GPIOL_55	Yes
25	GPIOL_52	Yes	26	GPIOL_53	Yes
27	GPIOL_50	Yes	28	GPIOL_51	Yes
29	GPIOL_48	Yes	30	GPIOL_49	Yes
31	GPIOL_46	Yes	32	GPIOL_47	Yes
33	GPIOL_44	Yes	34	GPIOL_45	Yes
35	GND	-	36	GND	-

(1) Dedicated PLL input pin that supplies the clock to FPGA's PLL.
(2) If you are using multi-image configuration, GPIOL_69 and GPIOL_70 are the CBSEL[1] and CBSEL[0] pins that select the image to use. For more information, refer to AN 006: Configuring Trion FPGAs

Table 6: H3 Pin Assignments

Pin Number	T20F256C Pin Name	DDIO Mode Supported	Pin Number	T20F256C Pin Name	DDIO Mode Supported
1	3V3	–	2	3V3	–
3	GPIOR_76[1]	No	4	GPIOR_77[1]	No
7	GPIOR_78	No	8	GPIOR_79	No
9	–	–	10	GPIOR_81	No
5	GPIOR_104	Yes	6	GPIOR_105	Yes
11	GPIOR_110	Yes	12	GPIOR_111	Yes
13	GPIOR_113	Yes	14	GPIOR_117	Yes
15	GPIOR_118	Yes	16	GPIOR_120[3]	Yes
17	GPIOR_121[3]	Yes	18	GPIOR_122[3]	Yes
19	GPIOR_123[3]	Yes	20	GPIOR_124[3]	Yes
21	GPIOR_126[3]	Yes	22	GPIOR_127[3]	Yes
23	GPIOR_149	Yes	24	GPIOR_150	Yes
25	GPIOR_151	Yes	26	GPIOR_153	Yes
27	GPIOR_154	Yes	28	GPIOR_155	Yes
29	GPIOR_156	Yes	30	GPIOR_158	Yes
31	GND	–	32	GND	–

[3] Dedicated clock input pin that supplies the clock to global clock network.

Table 7: H4 Pin Assignments

Pin Number	T20F256C Pin Name	DDIO Mode Supported	Pin Number	T20F256C Pin Name	DDIO Mode Supported
1	VCCIO1B_1C	-	2	VCCIO1B_1C	-
3	GPIOL_43	Yes	4	GPIOL_11	Yes
5	GPIOL_42	Yes	6	GPIOL_12	Yes
7	GPIOL_41	Yes	8	GPIOL_13	Yes
9	GPIOL_40	Yes	10	GPIOL_14	Yes
11	GPIOL_39	Yes	12	GPIOL_15	Yes
13	GPIOL_37	Yes	14	GPIOL_16	Yes
15	GPIOL_36	Yes	16	GPIOL_18	Yes
17	GPIOL_35	Yes	18	GPIOL_19	Yes
19	GPIOL_34	Yes	20	GPIOL_20	Yes
21	GPIOL_33	Yes	22	GPIOL_21	Yes
23	GPIOL_32	Yes	24	GPIOL_22	Yes
25	GPIOL_31(3)	Yes	26	GPIOL_23	Yes
27	GPIOL_30(3)	Yes	28	GPIOL_24(3)	Yes
29	GPIOL_29(3)	Yes	30	GPIOL_25(3)	Yes
31	GPIOL_28(3)	Yes	32	GPIOL_26(3)	Yes
33	GPIOL_27(3)	Yes	34	GND	-
35	GND	-	36	GND	-

Header H6

Header H6 is the JTAG interface. You can access the T20F256C JTAG pins through this header.

Pin Number	Signal Name	Description
1	3V3	3.3 V power supply
2	TDO	JTAG data output signal
3	TDI	JTAG data input signal
4	TCK	JTAG clock signal
5	FT_RST	Reset pin for on board FTDI FT2232 module (active low)
6	TMS	JTAG TMS mode select signal
7	GND	Ground
8	GND	Ground

Headers H7 and H8

H7 and H8 are 6-pin headers. You use a shunt across 2 pins to select 3.3 V, 2.5 V, or 1.8 V for the T20F256C bank VCCIO1B_1C and VCCIO1D_1E, respectively, from the on-board regulators.

By default, the board ships with a shunt connecting pins 5 and 6 to supply 3.3 V.

Table 8: Voltage Selection for VCCIO1B_1C and VCCIO1D_1E

Shunt	VCCIO1B_1C (H7)	VCCIO1D_1E (H8)
Connect pins 1 and 2	1.8 V	1.8 V
Connect pins 3 and 4	2.5 V	2.5 V
Connect pins 5 and 6	3.3 V (default)	3.3 V (default)

CAUTION: Only select one voltage at a time. Installing more than one shunt on H7 or H8 may cause contention.

Header J1

J1 is a 2-pin header that provides 5 V output as a power source for the external devices that interface with the development board. The 5 V DC power supply from header CON1 supplies this 5 V output.

Table 9: J1 Pin Assignments

Pin Number	Signal
1	5V
2	GND

Note: If you are supplying power to the board using the micro-USB cable only, limited power is available (<500 mA).

Headers J3 and J5

J3 is a 3-pin header used to select the source for the T20F256C clock input and PLL input. Drive a 3.3 V clock source input into the SMA connector, J5, if you are using the external clock source option.

Table 10: Clock Selection Pin Assignments

Pin Number	Signal
1	50 MHz on-board oscillator
2	GPIOR_125_CLK10 and GPIOR_157_PLLN
3	External clock source from SMA input J5

- A shunt connecting J3 pins 1 and 2 selects the 50 MHz on-board oscillator (default).
- A shunt connecting J3 pins 2 and 3 selects the clock source from SMA input J5.

User Outputs

The board has 8 green user LEDs that are connected to I/O pins in T20F256C banks 1A/1B. By default, the T20F256C I/O connected to these LEDs have a pull-up resistor that turns the LEDs off; to turn a given LED on, pull the corresponding I/O signal low.

Table 11: User Outputs

Reference Designator	T20F256C Pin Name	Active
D3	GPIOR_104	Low
D4	GPIOR_105	Low
D5	GPIOR_117	Low
D6	GPIOR_118	Low
D7	GPIOR_153	Low
D8	GPIOR_154	Low
D9	GPIOR_155	Low
D10	GPIOR_156	Low

User Inputs

The board has 3 pushbutton switches that you can use as inputs to the T20F256C device. The T20F256C bank 1A I/O signals connected to these switches have a pull-up resistor. When you press the switch, the signal drives low, indicating user input.

User Pushbuttons

Table 12: User Pushbuttons

Reference Designator	T20F256C Pin Name	Active
SW4	GPIOL_02	Low
SW5	GPIOL_04	Low
SW6	GPIOL_05	Low

User DIPswitches

Table 13: User DIPswitches

Reference Designator	T20F256C Pin Name	Off (Default)	On
SW3.1	GPIOR_128	Low	High
SW3.2	GPIOR_129	Low	High
SW3.3	GPIOR_130	Low	High

Installing Standoffs

Before using the board, attach the standoffs with the screws (M3 size) provided in the kit.

Warning: You can damage the board if you over tighten the screws. Tighten all screws to a torque between 4 ± 0.5 kgf/cm and 5 ± 0.5 kgf/cm.

Running the Demonstration Design

Efinix® preloads the Trion® T20 BGA256 Development Board with a demonstration design that operates the LEDs. The board receives power through USB cable. To run the design:

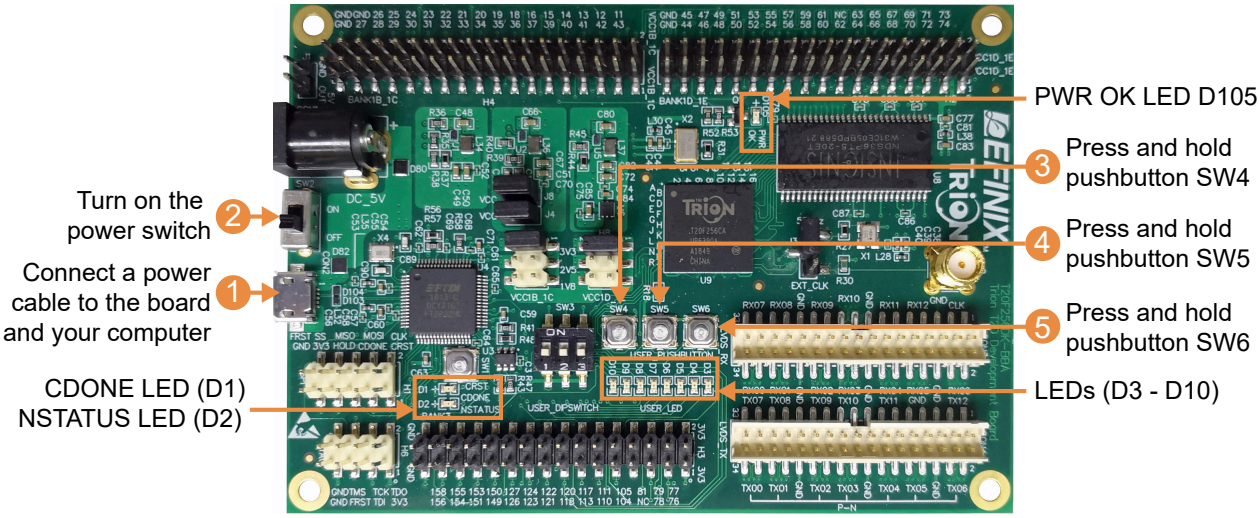
1. Connect the 5 V DC power source or USB cable to the board and to your computer.
2. Turn on the power switch (SW2). LED D105 (PWR OK) turns on, indicating that the board is receiving power correctly.

   Note: The Trion® T20 BGA256 Development Kit does not include a DC power adapter.

   - While configuration is in process, the configuration done LED D1 (CDONE) turns OFF to indicate the device is in configuration mode.
   - At the same time, the LED D2 (NSTATUS) turns ON to indicate there is no configuration error.
   - When configuration completes, the configuration done LED (D1) turns on again. Four green LEDs (D3 - D10) turn on, sweeping from D3 to D10 in ascending order.

   Note: If LED D105 does not turn off, the board is not receiving power correctly.
3. Press and hold pushbutton SW4. The LEDs blink alternately between group 1 (D3, D5, D7, and D9) and group 2 (D4, D6, D8, and D10).
4. Press and hold pushbutton SW5. The LEDs all blink on and off alternatively.
5. Press and hold pushbutton SW6. The LEDs sweep in backwards order from D10 to D3.

Figure 4: Running the Demonstration Design



Creating Your Own Design

The Trion® T20 BGA256 Development Board allows you to create and explore designs for the T20 device. Efinix® provides example code and designs to help you get started:

- Our Support Center includes examples targeting the board.
- The Efinity software includes also example designs that you can use as a starting point for your own project, and includes a step-by-step **tutorial**.
- **AN 027: Using the Raspberry Pi to HDMI Example Designs (T120 BGA576)** includes example designs with additional features for Trion® T20 BGA256 Development Board.

Appendix 1: Shared Resources

Some of the resources available on the Trion® T20 BGA256 Development Board are connected to more than one I/Os. You need to ensure there are no overlapping assignments when using these resources. The following table lists the resources shared by more than one I/Os. You can refer to this table to help you plan the resources available in the Trion® T20 BGA256 Development Board

Note: Resources that are not listed are only available from one I/O (see Headers on page 8).

Table 14: Trion® T20 BGA256 Development Board Shared Resources

<header name>.<pin name/number>

Resource	Connection 1	Connection 2
GPIOR_104	H3.5	LED.D3
GPIOR_105	H3.6	LED.D4
GPIOR_117	H3.14	LED.D5
GPIOR_118	H3.15	LED.D6
GPIOR_153	H3.26	LED.D7
GPIOR_154	H3.27	LED.D8
GPIOR_155	H3.28	LED.D9
GPIOR_156	H3.29	LED.D10

Revision History

Table 15: Revision History

Date	Version	Description
January 2025	1.9	**Note added** for variations in box content. (DOC-2340)
April 2023	1.8	Removed MIPI and LVDS Expansion Daughter Card section that was mistakenly added. Corrected board rev. to BBA.
February 2023	1.7	Added pointer to schematics for board version 3. (DOC-)
November 2022	1.6	Added link to AN 050: Managing Windows Drivers in Installing the Windows USB Drivers topic.
September 2022	1.5	Updated Installing Windows Driver.
June 2022	1.4	Added Appendix 1: Shared resources.
December 2021	1.3	Corrected the user DIPswitch pin names.
September 2021	1.2	Added USB driver installation topics. (DOC-463)
February 2021	1.1	Added note about referring to the power-up sequence in the data sheet when designing a board and recommending supervisor IC for `CRESET_N` (DOC-388).
February 2019	1.0	Initial release.